Answer the following groups

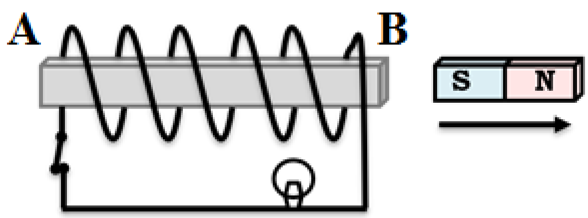
Group 1 :

Write the scientific term expressed by the following statements:

	Statement
1	The magnetic flux density which generates a force of 1 N on a straight wire of length 1 m carrying current of intensity 1 A placed perpendicular in a uniform magnetic field …………………………………………
2	Light Amplification by Stimulated Emission of Radiation …………………………………………
3	The law which states that the wavelength (λ_m) at maximum radiation intensity is inversely proportional to temperature on kelvin scale …………………………………………
4	The number of bonds broken per second will be equal to the number of bonds mended per second …………………………………………
5	The total number of magnetic flux lines passing through a surface and measured in weber …………………………………………
6	The ratio between output electric power produced from the secondary coil to the electric power given in the primary coil in the electric transformer …………………………………………
7	Emission of electrons from a metallic surface when the light energy falling on it …………………………………………
8	It is a phenomenon in which an induced electromotive force and also an induced current are generated in the conductor by changing magnetic field …………………………………………
9	Digital circuits that perform logic operations. …………………………………………
10	The point at which the total magnetic flux density vanishes …………………………………………

Group 2 :

Put circle around the correct answer

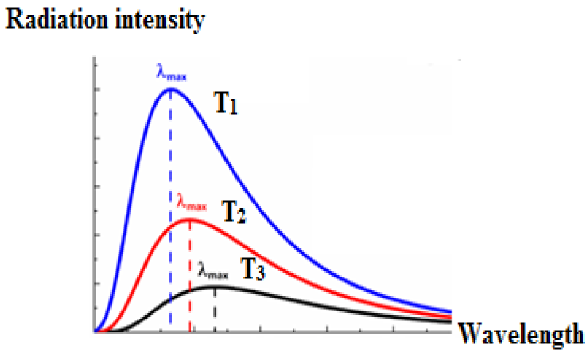
11	**The torque acting on a coil carrying current in a uniform magnetic field becomes zero when the plane of the coil is …………………. to the direction of the magnetic field.** a) perpendicular b) parallel c) inclined by an angle 30° d) inclined by an angle 60°
12	**When a coil rotates in a magnetic field, the direction of the induced electromotive force produced in the coil is changed each ……………cycle.** a) $\frac{1}{4}$ b) $\frac{1}{2}$ C) 1 d) $\frac{3}{4}$
13	**When the number of turns in the dynamo’s coil is doubled at constant other factors then the maximum electromotive force generated from it……………** a) Increases to the double b) decreases to half of its value c) remains constant d) Increases three times of its value
14	**In the opposite figure, when the magnet moves in the direction shown the pole formed at A is …**  a) North b) South c) west d) east
15	**When the temperature of pure silicon crystal (Si) rises, so its electrical conductivity.……** a) decreases b) does not change c) increases d) vanishes
16	**In Compton effect, when a photon of X-rays collides with a free electron initially at rest, the photon energy after collision …….** a) Increases b) decreases c) remain the same d) vanishes

17 **When emitted photons propagate having the same phase, it is said that they are**

a) Spreading
b) collimated
c) monochromatic
d) coherent

18 **The plots of intensity of radiation versus wave length of three black bodies at temperatures T_1, T_2 and T_3 are shown. Then**

a) $T_3 > T_2 > T_1$
b) $T_2 > T_3 > T_1$
c) $T_1 > T_3 > T_2$
d) $T_1 > T_2 > T_3$

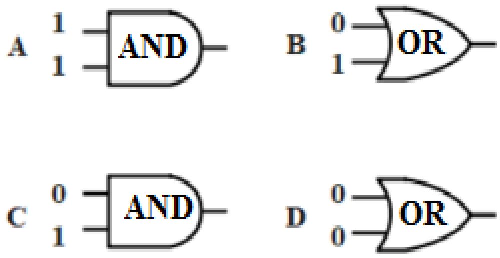


19 **Using x-rays in studying the crystalline structure of materials because it …**

a) Faster than light
b) Invisible
c) Diffract through atoms
d)ionizing radiation

20 **Which of the following gates its output is 1?**

a) Only B
b) Only D
c) Only A
d) A,B



Group 3:

Correct the underlined in the following statements

	Statement
21	Existence of **a repulsive** force between parallel wires of copper each carrying current in same direction ....
22	The electromotive force induced in a rectangular coil rotating in a magnetic field becomes zero when the coils' plane is **parallel** to magnetic field ....
23	Stimulated emission is dominant in **ordinary light** sources ....
24	Maximum induced emf in a straight wire moving **parallel** to a magnetic field ....
25	Emission of photoelectron with kinetic energy from metal occurs when the energy of incident radiation **smaller** than the work function of the metal ....
26	The minimum wavelength of the continuous radiation decreases when the potential difference between the cathode and the anode **decreases** ....
27	In the **reverse** connection the diode should have a small resistance hence a large current will flow through the diode ....
28	The coefficient of mutual inductance between two coils **increases** as the rate of change of current in the primary coil decreases ....
29	The emitter current is **smaller than** the sum of the collector current and the base current ....
30	Function of **Grid** in cathode ray tube (CRT) is heating up the cathode ....

Group 4:

Compare between each of the following

Ampere's right hand rule & Fleming's left hand rule in terms of the use of each

	Point of comparison	Ampere's right hand rule	Fleming's left hand rule
31 32	The use		

The physical relation used in calculating the value of the magnetic flux density at the center of a circular loop and at a point on the axis of a solenoid which the current traverse in each.

	Point of comparison	The magnetic flux density at the center of a circular loop	At point on the axis of a solenoid
33 34	The physical relation used		

Ordinary light source and laser with respect to monochromaticity

	Point of comparison	Ordinary light	LASER
35 36	Monochro-maticity		

P-type and N type semiconductors with respect to type of impurity

	Point of comparison	P-type semiconductor	N type semiconductor
37 38	type of impurity		

The continuous spectrum and the line spectrum of Spectrum of X-rays with respect to dependence on voltage between filament and target material

	Point of	The continuous spectrum	The line spectrum

	comparison		
39 **40**	**dependence on voltage between filament and target material**		

Group5:

Answer the following

	Problem
41 **42**	**Calculate the intensity of electric current** which passes through a circular coil of diameter 22 cm, 49 turns and produces a magnetic flux density in its center equal to 7×10^{-5} Tesla. ($\mu = 4\pi \times 10^{-7}$ Wb /A.m. ,$\pi = \frac{22}{7}$) ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ………………
43 **44**	An ideal step-down transformer is used to operate an electric lamp, which works on a potential difference of 30 volts. If an electric source of 240 volts is used & the number of tums of the primary coil is 480 turns, **calculate the number of tums of the secondary coils.** ……………………………………………………………………………………… ……………………………………………………………………………………… ………………………………………………………………………………………
45 **46**	An electric current of intensity 5 A flowing through a coil of self-inductance 0.001 H. If the current vanishes in 0.5 second, **calculate The induced electromotive force (emf) generated in the coil.** ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ………………………………………………………………………………………
47 **48**	npn transistor is used as a current amplifier, if the base current equals 1 mA and the current gain (β_e) is 200 calculate the collector current ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ……………………………………………………………………………………… ………………

49 **50**	A loop of cross-sectional area 0.1 m^2 and 100 turns, carries a current of intensity 20 A is placed parallel to a uniform magnetic field of flux density 0.5 T. Calculate the torque acting on the loop. ……………………………………………………………………………………… ………………………………………………………………………………..………… ……………………………………………………………………………………… ………………………………………………………………

انتهت الأسئلة